TASK SPECIFICATION: P5-02 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

Parking Spots & Zones Real-Time Query API

Git Branch: `feature/P5-02-spots-api` · Target Pull Request: feature/P5-02-spots-api

Core Objective & Technical Scope:
High-performance read API delivering real-time spot occupancy status with zone, status, and accessibility filtering.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-02-spots-api`
Upstream Dependencies	P5-01		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Implement SpotService with optimized projection queries for rapid grid rendering.
2. Create SpotController exposing /api/v1/parking/spots with filter parameters.
3. Implement zone aggregate statistics endpoint (total, available, occupied).
4. Author API integration tests.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `parking/api/*`
- `parking/application/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Accurate status filtering (Available, Occupied, Reserved, Maintenance).
- Sub-50ms API response time under concurrent read load.

VERIFICATION & TEST COVERAGE STRATEGY

- SpotServiceTest
- SpotControllerMockMvcTest